

## What is perceptron?

A perceptron is a machine learning algorithm that mimics the way a neuron in the brain works. It's also known as a single-layer neural network, or an artificial neuron.

A perceptron's function is to detect patterns or features in input data. It's used for supervised learning of binary classifiers.

**The four components of a perceptron are:**

- Input layer: Receives input data
- Weighted sum: Combines input values with weights
- Activation function: Determines output based on weighted sum
- Output layer: Produces classification result

A single layer perceptron (SLP) is the simplest type of artificial neural network. It can only classify linearly separable cases with a binary target (1, 0).

The Perceptron Convergence Theorem states that if there exists a set of weights that are amenable to treatment with Perceptron, then the Perceptron learning algorithm will converge

## What is single layer perceptron?

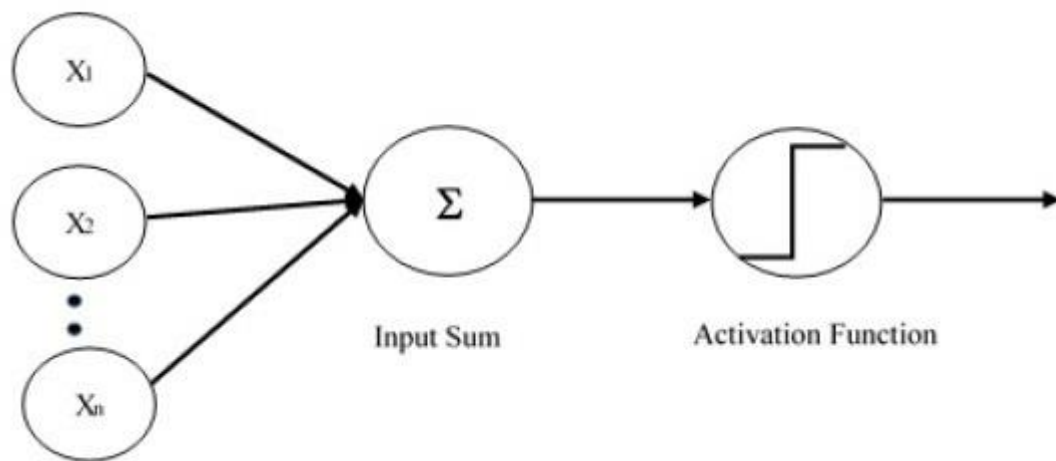
A single layer perceptron (SLP) is a type of artificial neural network that's based on a threshold transfer function. It's the simplest type of feed-forward network and doesn't have any hidden layers.

**SLPs have the following advantages:**

- Easy to implement
- Easy to train
- Easy to understand
- Clear geometric interpretation
- Perform well on linearly separable problems

SLPs can only classify linearly separable cases with a binary target. They can perform well on problems such as: Logical operations, Linear regression, Binary classification.

SLPs were proposed by Frank Rosenblatt in 1958.



## What is multi layer perceptron?

A multilayer perceptron (MLP) is a type of artificial neural network that is used for classification and regression tasks in machine learning. It is a fully connected multi-layer neural network that consists of three types of layers:

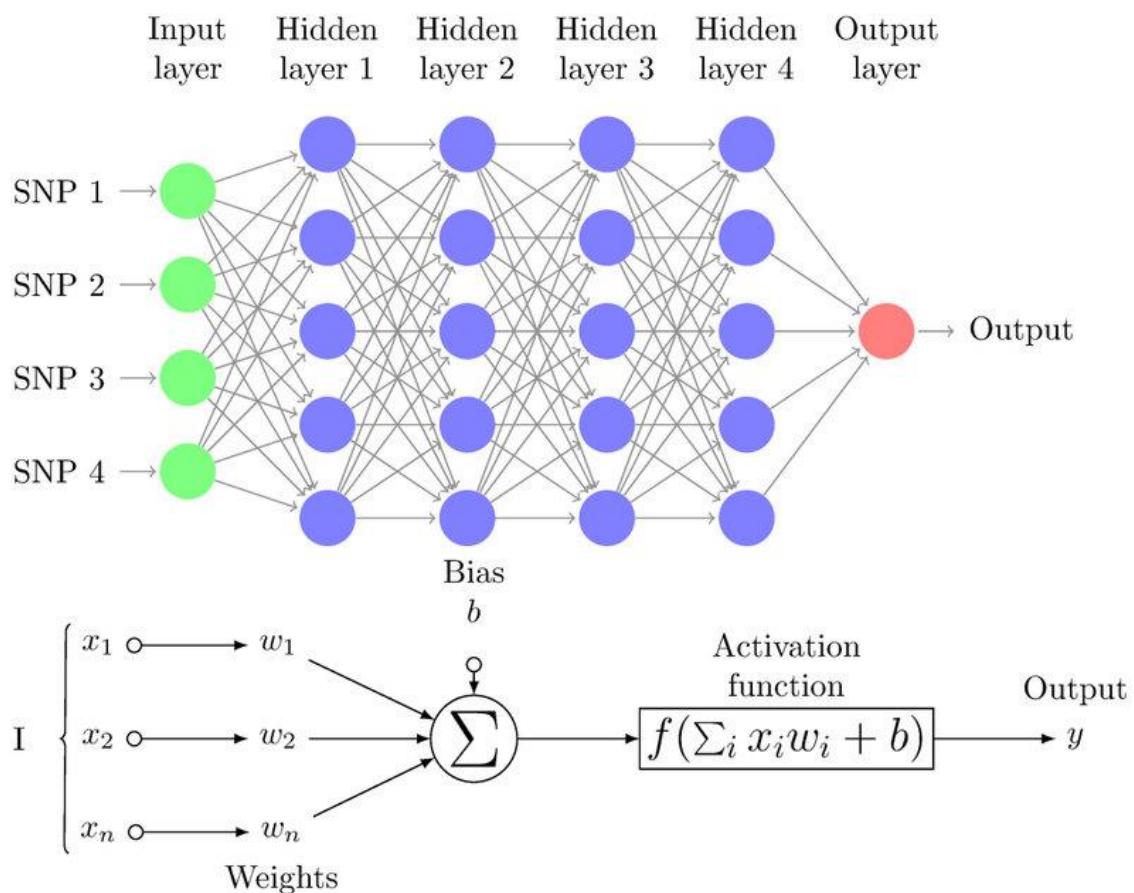
- Input layer: Receives the input signal to be processed
- Hidden layer: Can have one or more layers
- Output layer: Transforms the output of the hidden layer via an activation function

MLP neurons can use any arbitrary activation function. Some commonly used activation functions include: ReLU, Sigmoid, Tanh.

MLP is a supervised learning algorithm that learns a function by training on a dataset. It uses a Backpropagation algorithm to increase the accuracy of the training model.

**Steps to implement a multilayer perceptron model include:**

1. Open Google Colab notebook
2. Import libraries and modules
3. Choose or download a dataset
4. Turn pixels into floating-point values
5. Visualize the data
6. Make input, hidden, and output layers
7. Compile the model



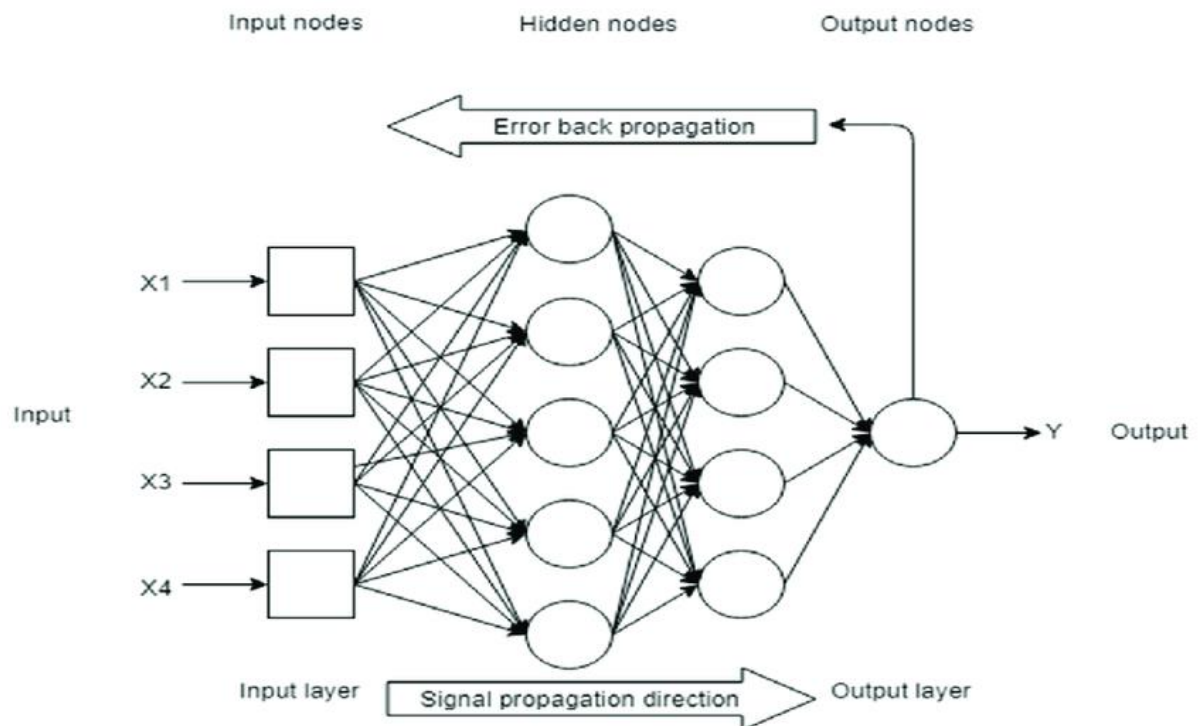
## What is backpropagation?

Backpropagation is a method for training neural networks. It involves adjusting the weights of a neural network by analyzing the error rate from the previous iteration. Backpropagation is a type of automatic differentiation algorithm.

**Here are some steps for backpropagation:**

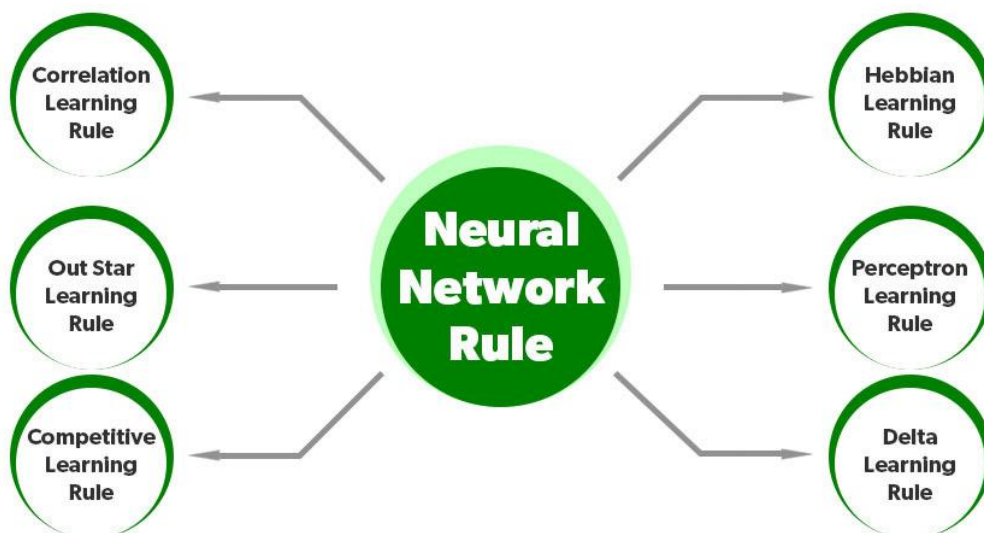
1. **Inputs:** Inputs arrive through the preconnected path.
2. **Modeling:** The input is modeled using true weights.
3. **Output:** Calculate the output of each neuron from the input layer to the hidden layer to the output layer.
4. **Error:** Compare the output with the target values and forecast the error in the outputs.
5. **Weights:** Calculate and update the weights of the previous inputs based on the error.

Backpropagation is the essence of neural network training. It can make a neural network more reliable by increasing its generalization.



## Learning rules in ANN?

A learning rule in a neural network is a mathematical logic or algorithm that improves the network's performance. It's a set of instructions that helps reinforce a pattern.



Learning rules update the weights and bias levels of a network when it simulates in a specific data environment. This is an iterative process that helps the neural network learn from the prevailing conditions.

There are six learning rules that are widely used by neural networks for training. The three main types of learning in neural networks are: Supervised learning, Unsupervised learning, Reinforcement learning.

One type of learning rule is the Winner-takes-All rule. In this rule, all the output nodes compete with each other to represent the input pattern. The winner is the node with the most outputs and is given the output 1.